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DSC 640: Final

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Final: Proposing New Child Care Centers in Pennsylvania

Summary of Analysis

Findings

Through data exploration and studying the technical guide, I realized that data quality varied by state. Since my plan for the story was going to be studying the change in prices by county over time, I needed a state that was not imputing data too often and had fairly detailed county-level data. For this exercise, I can imagine a company deciding to enter into a particular state. Since Pennsylvania has very complete data, I filtered on that state for my project.

Using Tableau, I began exploring the median income by county, and the cost of childcare. I used calculated fields to build a “Percent of household income needed for childcare” and used a parameter so I could toggle between “family-based” and “center-based”.

I could see that center-based childcare was more expensive than family-based childcare. I built line graphs to chart the change in price over time and found that center-based childcare is increasing in price faster than family-based. By using Tableau’s Trend Line functionality, I could find the change in dollars per year. For instance:

```
Avg. MC Preschool = 0.0077377*Year of Study Year + -174.967  
R-Squared: 0.994978  
P-value: < 0.0001
```

The slope of the center-based preschool prices is .0077377. Tableau’s time is daily, so if I multiply by 365, I see that center-based preschool weekly prices are increasing by about \$2.82 each year.

Using the same process for family-based:

```
Avg. MFCC Preschool = 0.00440987*Year of Study Year + -56.6397  
R-Squared: 0.972519  
P-value: < 0.0001
```

$$.00440987 * 365 = \$1.61$$

So not only is the center-based childcare more expensive, but it's also growing in cost faster.

Taking it a step further, I looked specifically at one of the counties that has the highest 'percent of income needed for childcare' scores: Mercer County.

```
Avg. MC Preschool = 0.0161705*Year of Study Year + -504.46
R-Squared: 0.864449
P-value: < 0.0001
```

$.0161705 * 365 = \$5.90$. It's no wonder that Mercer County is paying so much of their income towards childcare, when weekly prices are increasing that much.

I improved these "by-hand" calculations of trend above in Milestone 3, using fixed level of detail calculations in Tableau.

MC Average Trend Slope

```
(
  { FIXED : SUM([YearNum] * [MC Average]) }
  - ({ FIXED : SUM([YearNum]) } * { FIXED : SUM([MC Average]) }) / { FIXED : COUNT([YearNum]) }
)
/
(
  { FIXED : SUM([YearNum]^2) }
  - ({ FIXED : SUM([YearNum]) }^2 / { FIXED : COUNT([YearNum]) })
)
```

This slope calculation gives the average weekly increase in price, which can be multiplied by 52 to give an average annual increase.

Assumptions

The technical guide gives details on the assumptions, interpolations and imputing that needed to be done to collect the data. Filtering on Pennsylvania data alleviates some of those concerns, however, with less interpolation and better county-level data than what other states provided.

Items Clarified in Final Presentation

Feedback from Milestone 3 suggested that my dashboard could use more information, so I added a "County at a Glance" section that gives employment data for the county to help management understand the population of the county.

Direction of Story

I will imagine that my company builds franchise childcare locations, and that we are looking to expand into the Pennsylvania market. I'll be able to use this data to find counties where center-based prices have grown high and are particularly challenging given the income levels in the county. My analysis showed that Philadelphia, Mercer and Centre counties had the best opportunity, because their median cost of childcare represented the largest percentage of medium income in that county.

Target Audience

My primary audience will be the management level at my hypothetical company that will need to approve the new locations. They'll be aware of the market and the measurements I'm using, but they will likely need a refresher on the more complicated calculations.

I'll also prepare media for all company employees to announce our company's plan for new locations. This will be much less technical and geared towards generating excitement.

Medium Choice

First, I will produce a dashboard that will allow my teams to explore Pennsylvania and get details on the specific regions discussed in my proposal. My proposal will be in the form of a PowerPoint presentation that will specifically name the counties I propose for new locations. Finally, I'll create an infographic to use for our internal announcement. It will call out the headlines for the proposal and encourage the team as we enter the new market.

Design Decisions

My dashboard and corresponding PowerPoint use a simple color palette:

- Red highlights the name of a selected county
- Shades of blue show the varying degrees of the measurements.

This simplified palette draws attention to the selected county, so that it's clear when the selections have changed.

I also make sure that the graphics I use to denote “Infant”, “Toddler” and “Preschool” are the same in both the presentation and the dashboard, so that there is a consistent visual connection.

The Infographic poster uses a slightly different color palette: blue and magenta. There was no need to use the same colors as the presentation, because these are different audiences. Red often denotes ‘error’ or ‘stop’. I wanted the infographic to be inviting, so I changed it to more of a magenta. However, I’m using blue to represent our company, and the magenta, which can still share some of red’s connotations, to represent things we want to categorize as ‘less than optimal’, like the high % of income that goes to childcare, or the high prices in those counties. Through this color choice, I hope our employees get the idea that our company can fix the childcare struggles that people in these counties face.

Ethical Considerations

Further research should be done on cost analysis. For the purposes of this assignment, I’m assuming that our company can effectively provide a high standard of care at a rate of 10% off the median in the county. It could be that the distribution of childcare between affluent and struggling neighborhoods within the county causes the median values to be inflated, and our ‘10% off’ price might appear good on paper but wouldn’t actually help the neediest communities.

For the purposes of this assignment, I assumed that the data sources provided were accurate and correct. But having read through the data collection process, I would potentially want to do some of my own market analysis into the prices to confirm, rather than using this one single source.

Lessons Learned

Throughout my progress towards this degree, I’ve found that my skills in visualization are more on the ‘reporting’ side, and less on the ‘analytical’ side. It was difficult to create the trend calculation, and I’m realizing that I often lean more on ‘telling the facts’ instead of ‘analyzing for insights’. I’m realizing that I need to push myself to ask analytical questions that may be harder to answer than by what’s simply available in the data. There are potentially some other analytical questions I could explore with this data set, if I were doing this project again.

I most enjoy the dashboard creation part. Once I get the data cleaned and ready and I can start playing in Tableau, where I’m comfortable and experienced, I get a lot of create momentum.